

Conditional Independence in DAGs

STSCI/INFO/ILRST 3900: Causal Inference

24 Sep 2026

Learning goals for today

At the end of class, you will be able to:

1. Identify whether paths in a causal diagram are open or blocked given a conditioning set
2. Understand why conditioning on colliders differs from conditioning on non-colliders

Logistics

- ▶ Project teams assigned yesterday in discussion section
- ▶ Part I of project due end of October

Causal Graphs

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- ▶ Causal Directed Acyclic Graphs (DAG) help communicate modeling assumptions and implications
- ▶ Path is a sequence of connected nodes which does not contain repeats
- ▶ **Causal path** path in which all arrows point away from the treatment toward the outcome
- ▶ Check (marginal) independence by looking at paths in graph
- ▶ Check (marginal) exchangeability by looking at paths in graph

Checking Marginal Independence

$$A \rightarrow Z_1 \rightarrow Z_2 \leftarrow Z_3 \rightarrow Y$$

- ▶ Two types of nodes on a path:
 - ▶ Collider: $\rightarrow Z \leftarrow$

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- ▶ Path is unblocked if it does **not** contain a collider
- ▶ Two variables are statistically dependant if there is an unblocked path between them

Exchangeability and DAGs

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- ▶ Conditional Exchangeability holds **given** L if all unblocked paths between A and Y are causal paths

Exchangeability and DAGs

- ▶ List out all paths from A to Y
- ▶ Exchangeability holds if all unblocked paths are causal paths
- ▶ **Today:** How do we check for conditional exchangeability?
- ▶ Conditional Exchangeability holds **given** L if all unblocked paths between A and Y are causal paths
- ▶ How do we tell if a path is blocked/unblocked conditional on L ?

Open or blocked?

How do we check if a path in the DAG is open or blocked when conditioning on a set of variables L ?

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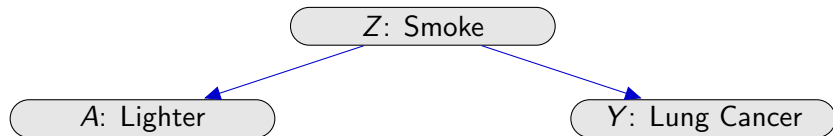
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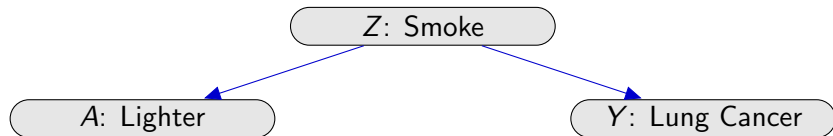
- ▶ If **any** node on the path is blocked, then the entire path is blocked
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Two variables are dependent conditional on L if there is an unblocked path (when conditioning on L) between them

Common cause



Common cause



If Z has a causal effect on both A and Y , the path is blocked when we condition on Z

Mediation



Mediation



If A effects Y through Z , the path is blocked when we condition on Z

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For non-colliders

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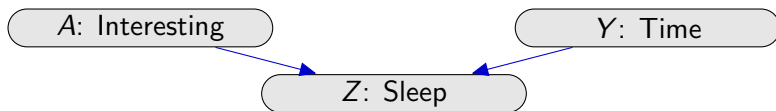
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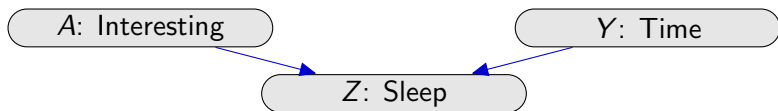
Rule:

- ▶ If Z is in the conditioning set, then Z is blocked
- ▶ Otherwise, Z is open

Collider



Collider

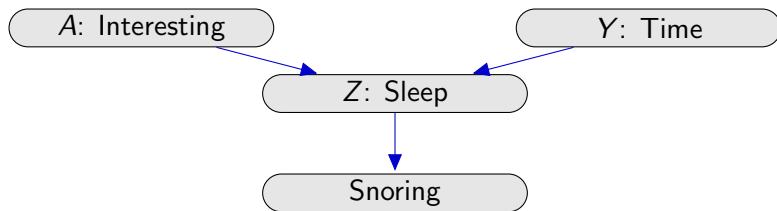


Mathematically,

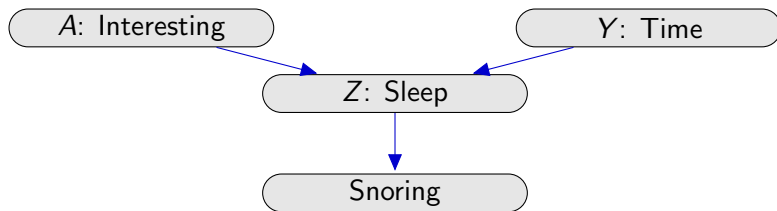
$$Z = X + Y$$

If we keep Z fixed, but increase X , then to preserve the equation, Y must decrease

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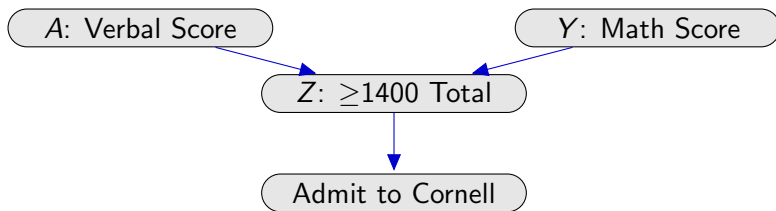


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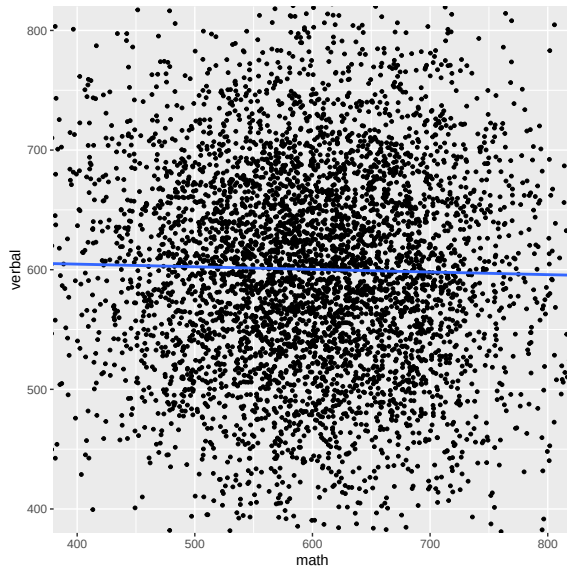


- If there is a causal path $X \rightarrow \dots \rightarrow Z$, then Z is a descendant of X

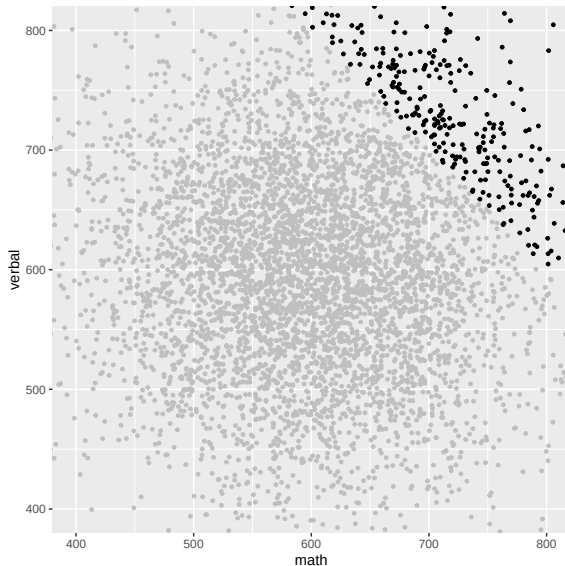
Collider bias: SAT scores



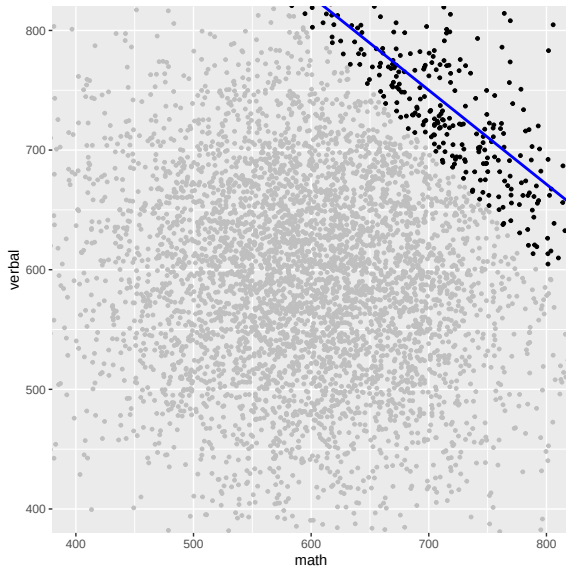
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How to check if a node is open or blocked:

- ▶ If non-collider:
 - ▶ Open if it is not in the conditioning set
 - ▶ Blocked if it is in the conditioning set

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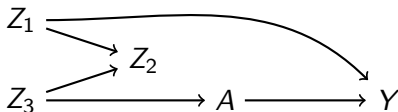
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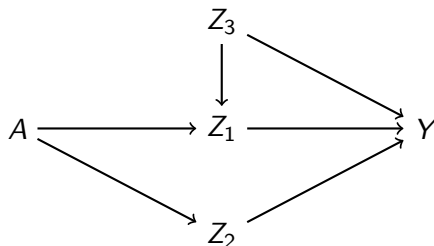
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Exercise



- ▶ What are the paths from A to Y ?
- ▶ When conditioning on $L = \{Z_1\}$ are those paths open or blocked?
- ▶ When conditioning on $L = \{Z_2\}$ are those paths open or blocked?
- ▶ When conditioning $L = \{Z_1, Z_2\}$ are those paths open or blocked?

Exercise



- What are the paths from A to Y ?
- When conditioning on $L = \{Z_2\}$ are those paths open or blocked?
- When conditioning $L = \{Z_1, Z_2\}$ are those paths open or blocked?

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